



Tyler C. McCann

Technical Lead, Red Team Operator
(OSCE³)

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ACHIEVEMENTS

Content within the OSCP (OffSec)

My SigmaPotato project is taught in the official OSCP course material as of Q4 2024.

Cyber Apocalypse CTF 2023 (HackTheBox)

Team placed in the top 3% globally (245/6483).

EDUCATION

Joint Cyber Analysis Course (JCAC)

1000+ hour NSA accredited course that teaches the baseline knowledge and skills required for offensive and defensive cyber operations.

December 2018 – June 2019

Middle Tennessee State University (MTSU)

Computer Engineering, B.S.
(In-Progress)

August 2016 – May 2018

EXPERIENCE

Active Security Clearance:
TS/SCI w/ CI Polygraph

Red Team Operator

Associate Director | UBS

● Technical & Engagement Lead | November 2024 – Present

Led internal Red and collaborative Purple Team engagements with a focus on Azure security, conducting sophisticated attacks while maintaining cloud-based attacker infrastructure. Managed multiple C2 frameworks and evasive payload delivery systems, developing custom tooling to enhance tradecraft, bypass EDR, and assess defensive controls. Ensured engagements aligned with governance standards, controlling operational risk.

Outflank C2

Cobalt Strike

Microsoft Azure

EDR Evasion

Intel x64

DoD SkillBridge Internship

SkillBridge Intern | SIXGEN

● Exploit Developer | March 2024 – June 2024

Developed proficiency in Windows User Level exploits for Intel x86 architectures and exploit crafting techniques, bypassing common security mitigations such as DEP and ALSR. Other topics covered include binary obfuscation for endpoint AV evasion.

Intel x86

Assembly

Reverse Engineering

Win32 API

Cyber Operations Specialist (17C)

Staff Sergeant (SSG) | U.S. Army Cyber Protection Brigade

● Cyber Threat Emulator | June 2020 – September 2024

Senior lead within the unit's Cyber Threat Emulation (CTE) cell, mentoring and developing junior offensive operators while specializing in conducting custom Red and Purple Team exercises against the brigade's Cyber Protection Teams (CPTs), emulating Advanced Persistent Threats (APTs). Additionally, generated analytics by virtualizing, deploying, and exploiting CVEs and TTPs across critical infrastructure environments, ranging from industrial control systems (ICS) to enterprise networks.

Red Team

Active Directory

ICS

AV Evasion

Web Exploitation

SQL

DevOps

● Analytic Support Officer | February 2020 – June 2020

Subject Matter Expert (SME) for both detecting information gaps and generating potential analytics for priority information requirements (PIRs) for both upcoming and ongoing missions.

Data Analytics

Dashboards

Digital Forensics

● Network Analyst | August 2018 – February 2020

Senior network analyst on a U.S. European Command (USEUCOM) based Cyber Protection Team (CPT), engaging in a variety of mission types from incident response to network hardening and threat hunting. During this time, advised and assisted junior analysts during two forward operations in Europe. Gained proficiency in Splunk, Elastic Stack, and Endgame EDR.

Linux

Windows

Security Onion

Splunk

Elastic Stack

ESXi

Sysmon

SKILLS

- Intel x86
- Assembly
- DevOps
- Networking
- CI/CD
- Github Pages
- Proxmox
- ESXi
- Automation
- Scripting
- Network TAPs
- SIEM
- Dashboards
- Data Analytics
- Digital Forensics
- Security Onion
- Splunk
- Elastic Stack
- Sysmon
- Linux
- Shellcode
- ROP
- PenTesting
- Active Directory
- AV Evasion
- IEEE 802.11
- Web Exploitation
- SQL
- PrivEsc
- Reverse Shells
- XXE
- XSS
- Template Injection
- Exfiltration
- Bash
- PowerShell
- C#
- .NET
- Microsoft Office
- Windows

CERTIFICATIONS

DoD Directive 8570 Compliance:
IAT-III, CSSP Analyst, CSSP Infrastructure Support,
CSSP Incident Responder, CSSP Auditor

- OffSec Certified Expert 3 (OSCE³) May 2024 | OffSec
- OffSec Exploit Developer (OSD) May 2024 | OffSec
- OffSec Experienced Penetration Tester (OSEP) April 2023 | OffSec
- OffSec Certified Professional (OSCP) March 2023 | OffSec
- OffSec Web Expert (OSWE) March 2024 | OffSec
- OffSec Web Assessor (OSWA) December 2023 | OffSec
- OffSec Wireless Professional (OSWP) April 2023 | OffSec
- Red Team Apprentice Certified (RTAC) March 2024 | k>fivefour
- Certified Ethical Hacker (CEH) June 2024 | EC-Council
- GIAC Certified Enterprise Defender (GCED) August 2020 | SANS Institute
- OffSec Defense Analyst (OSDA) May 2024 | OffSec
- Microsoft Azure Fundamentals (AZ-900) January 2025 | Microsoft

PROJECTS

SigmaPotato

<https://github.com/tylerdotrar/SigmaPotato>

Selmpersonate privilege escalation tool for Windows 8 - 11 and Windows Server 2012 - 2022 with extensive PowerShell and .NET reflection support.

C#

.NET

Privilege Escalation

ShellcodeLoaderPS

<https://github.com/tylerdotrar/ShellcodeLoaderPS>

Educational PowerShell library containing numerous shellcode injection techniques, with examples demonstrating the Win32 API, Native API, and both direct and indirect syscalls. Written entirely in pure PowerShell without embedded C# code.

PowerShell

Shellcode Injection

Win32 API

Native API

In/Direct Syscalls

RGBwiki

<https://rgbwiki.com>

Github Pages hosted wiki website designed around offensive TTPs (Red), DevOps deployment (Green), and defensive cyber knowledge (Blue).

CI/CD

Github Pages

MkDocs

Documentation

genrev

<https://github.com/tylerdotrar/genrev>

Modular Python tool that uses the Python keystone-engine library to convert Intel (x86) assembly instructions into 32-bit Windows reverse shell shellcode.

Python

Intel x86

Assembly

Shellcode